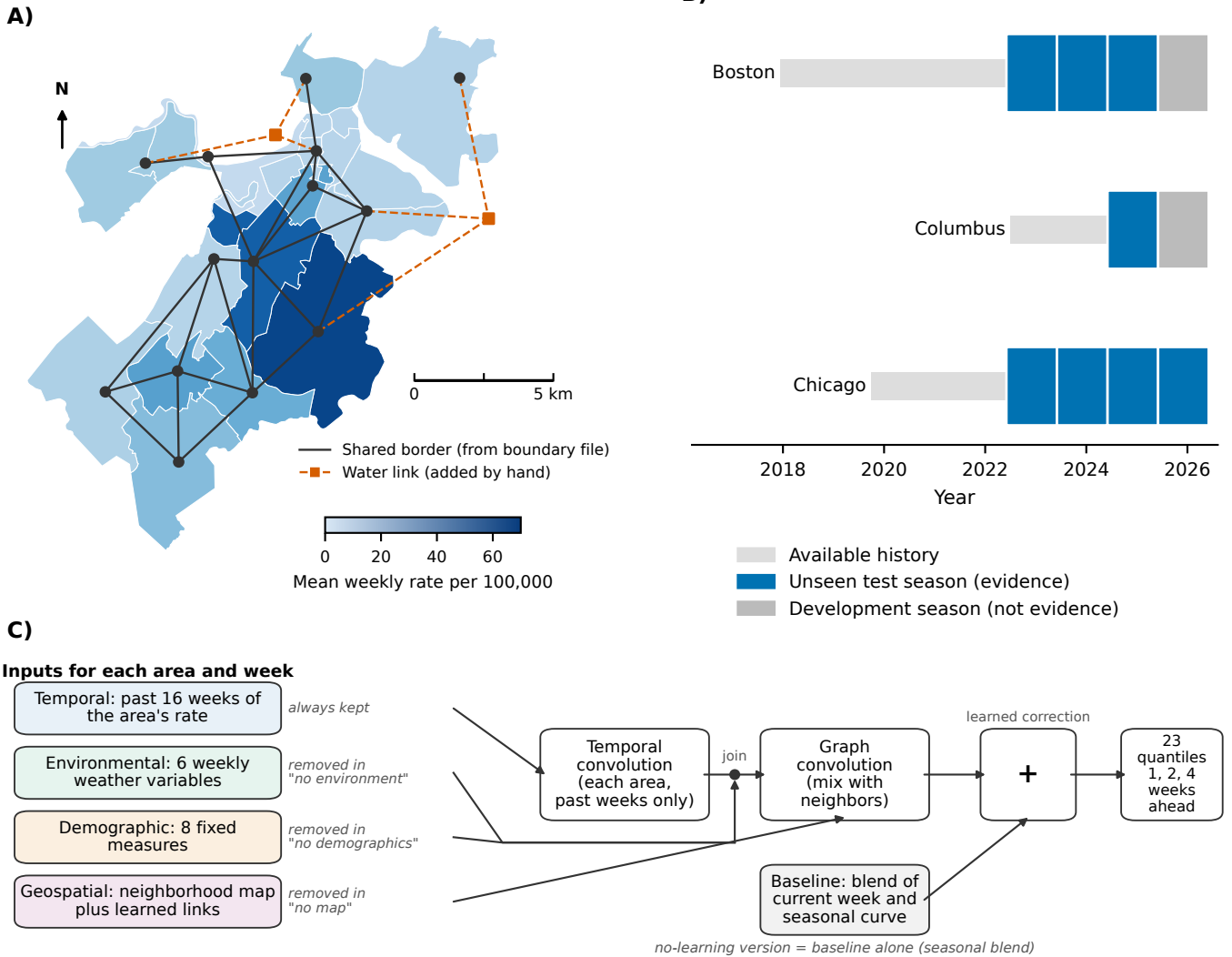

A)
N
0
5 km
Shared border (from boundary file)
Water link (added by hand)
0
20
40
60
Mean weekly rate per 100,000
B)
Boston
Columbus
Chicago
2018
2020
2022
2024
2026
Year
Available history
Unseen test season (evidence)
Development season (not evidence)
C)
Inputs for each area and week
Temporal: past 16 weeks of the area's rate
always kept
Environmental: 6 weekly weather variables
removed in "no environment"
Demographic: 8 fixed measures
removed in "no demographics"
Geospatial: neighborhood map plus learned links
removed in "no map"
Temporal convolution (each area, past weeks only)
join
Graph convolution (mix with neighbors)
learned correction
+
23 quantiles 1, 2, 4 weeks ahead
Baseline: blend of current week and seasonal curve
no-learning version = baseline alone (seasonal blend)